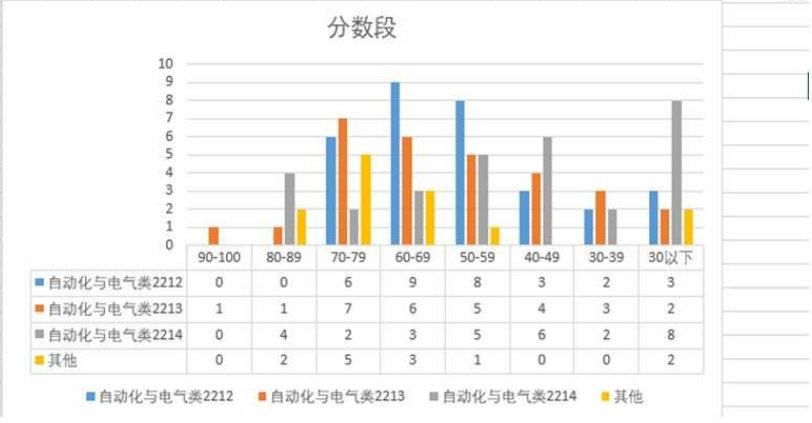
第8章 耦合电路

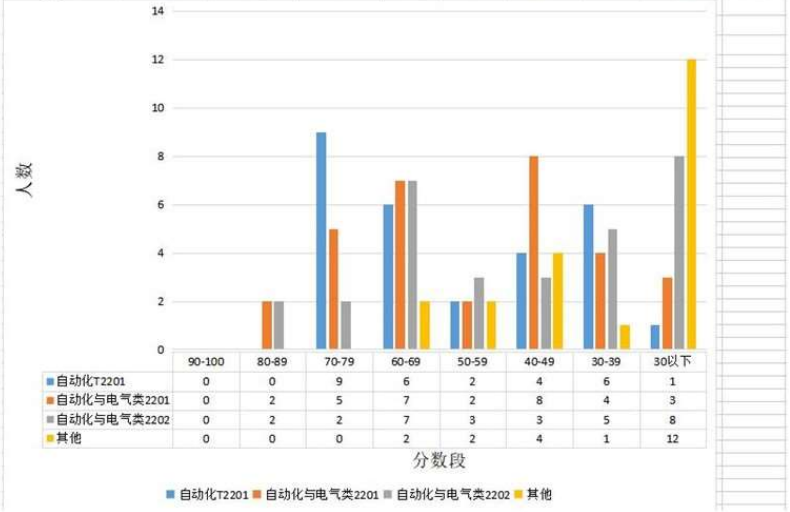
8.1 含耦合电感的正弦电路

8.2 含耦合电感电路的计算

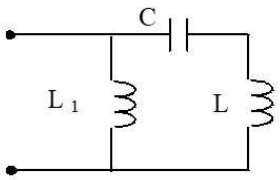
	90-100	80-89	70-79	60-69	50-59	40-49	30-39	30以下	最高分
自动化与电气类2212	0	0	6	9	8	3	2	3	77
自动化与电气类2213	1	1	7	6	5	4	3	2	91
自动化与电气类2214	0	4	2	3	5	6	2	8	86
其他	0	2	5	3	1	0	0	2	83



	90-100	80-89	70-79	60-69	50-59	40-49	30-39	30以下	最高分
自动化T2201	0	0	9	6	2	4	6	1	78
自动化与电气	0	2	5	7	2	8	4	3	82
自动化与电气	0	2	2	7	3	3	5	8	83
其他	0	0	0	2	2	4	1	12	69



三、由纯电感和纯电容构成的谐振电路



串联谐振 ω_0　　$\omega_0 = \dfrac{1}{\sqrt{LC}}$

并联谐振 ω_1

$$Y(\omega_1) = \frac{1}{j\omega_1 L_1} + \frac{1}{j(\omega_1 L - \frac{1}{\omega_1 C})}$$

$$\omega_1 = \frac{1}{\sqrt{(L_1 + L)C}}$$

问：$\omega_0 > \omega_1$？　$\omega_0 < \omega_1$？

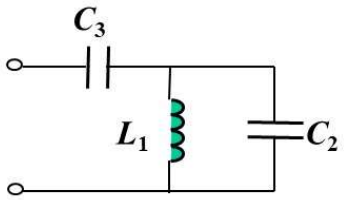
注意：

$Im(Z_{in})=0$，串联谐振；相当于短路。

$Im(Y_{in})=0$，并联谐振。相当于开路。

2023年5月6日

例：求谐振角频率。



$$Z(\omega)=\frac{1}{\mathrm{j}\omega C_3}+\frac{\mathrm{j}\omega L_1\dfrac{1}{\mathrm{j}\omega C_2}}{\mathrm{j}\omega L_1+\dfrac{1}{\mathrm{j}\omega C_2}}$$

$$=\frac{1}{\mathrm{j}\omega C_3}+\frac{\mathrm{j}\omega L_1}{1-\omega^2 L_1 C_2}$$

$$=-\mathrm{j}\frac{1-\omega^2 L_1(C_2+C_3)}{\omega C_3(1-\omega^2 L_1 C_2)}$$

分别令分子、分母为零，可得：

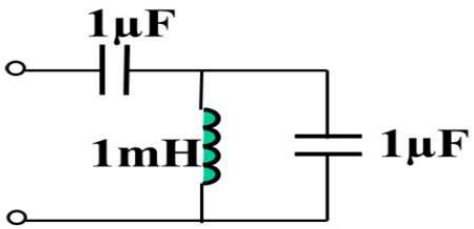
$$\omega_1=\frac{1}{\sqrt{L_1(C_2+C_3)}}$$ 发生串联谐振 $Z_0=0$

$$\omega_2=\frac{1}{\sqrt{L_1 C_2}}$$ 发生并联谐振 $Z_0=\infty$

2023年5月6日

单选题 1分

对于图示电路，当频率为何值时，会发生串联谐振？



A 3.56 kHz

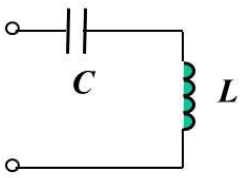
B 5.03 kHz

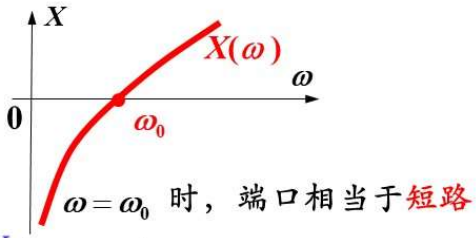
C 22.4 kHz

D 31.6 kHz

2023年5月6日

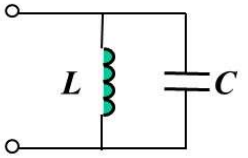
LC串联谐振

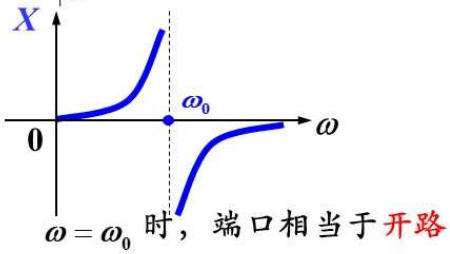




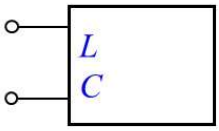
$\omega=\omega_0$ 时，端口相当于短路

LC并联谐振





$\omega=\omega_0$ 时，端口相当于开路

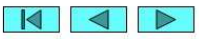


$$Z(\omega)=\mathrm{j}\frac{f_1(\omega)}{f_2(\omega)}$$

$f_1(\omega_0)=0$ 时，电路发生串联谐振

$f_2(\omega_0)=0$ 时，电路发生并联谐振

2023年5月6日



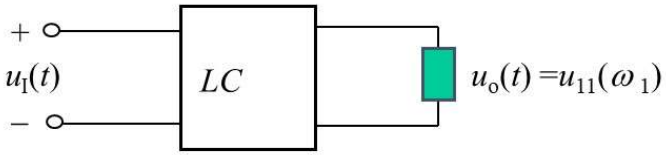
思考：

激励 $u_1(t)$，包含两个频率ω_1、 ω_2分量 ($\omega_1<\omega_2$)：

$$u_1(t)=u_{11}(\omega_1)+u_{12}(\omega_2)$$

要求负载电压$u_o(t)$只有$u_{11}(\omega_1)$频率电压，（无ω_2频率电压）。

如何实现?



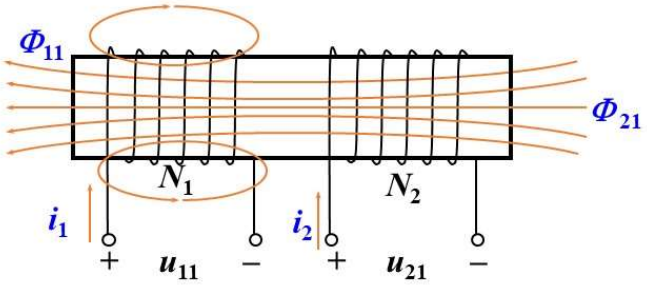
若$\omega_1>\omega_2$，仍要只得到ω_1频率电压，如何设计电路?

2023年5月6日



8-1 含耦合电感的正弦电路

互感电压



当i_1、u_{11}、u_{21}方向与Φ 符合右手螺旋时，

根据电磁感应定律和楞次定律：

$$u_1 = u_{11} + u_{12} = L_1\frac{\mathrm{d}i_1}{\mathrm{d}t} + M\frac{\mathrm{d}i_2}{\mathrm{d}t}$$

$$u_2 = u_{21} + u_{22} = M\frac{\mathrm{d}i_1}{\mathrm{d}t} + L_2\frac{\mathrm{d}i_2}{\mathrm{d}t}$$

自感电压

互感电压

$$u_{11} = \frac{\mathrm{d}\Psi_{11}}{\mathrm{d}t} = L_1\frac{\mathrm{d}i_1}{\mathrm{d}t} \qquad u_{12} = \frac{\mathrm{d}\Psi_{12}}{\mathrm{d}t} = M_{12}\frac{\mathrm{d}i_2}{\mathrm{d}t}$$

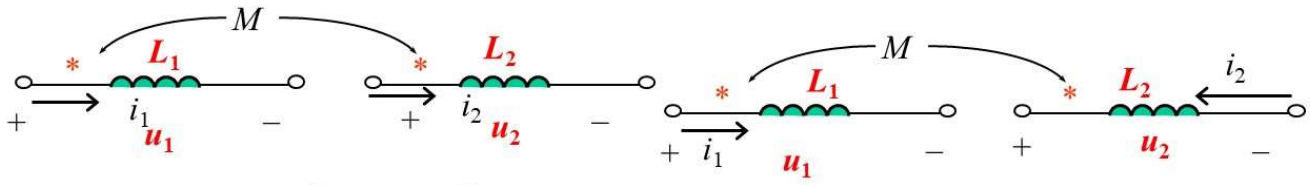
同理：

$$u_{21} = \frac{\mathrm{d}\Psi_{21}}{\mathrm{d}t} = M_{21}\frac{\mathrm{d}i_1}{\mathrm{d}t} \qquad u_{22} = \frac{\mathrm{d}\Psi_{22}}{\mathrm{d}t} = L_2\frac{\mathrm{d}i_2}{\mathrm{d}t}$$

三、同名端

互感电压的方向与线圈有关，但实际的互感元件制成后是密封的。

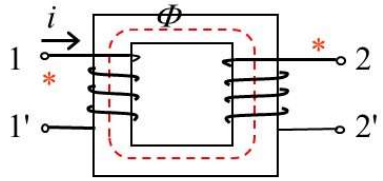
因此，为了表示自感磁通与互感磁通是否相助，采用同名端的标记方法：用“*”，“.”

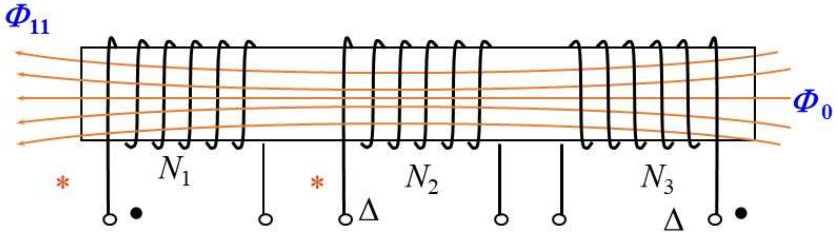


$$u_1 = L_1\frac{di_1}{dt} + M\frac{di_2}{dt} \qquad u_1 = L_1\frac{di_1}{dt} - M\frac{di_2}{dt}$$

$$u_2 = L_2\frac{di_2}{dt} + M\frac{di_1}{dt} \qquad u_2 = -L_2\frac{di_2}{dt} + M\frac{di_1}{dt}$$

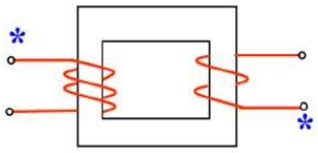
例1





单选题 1分

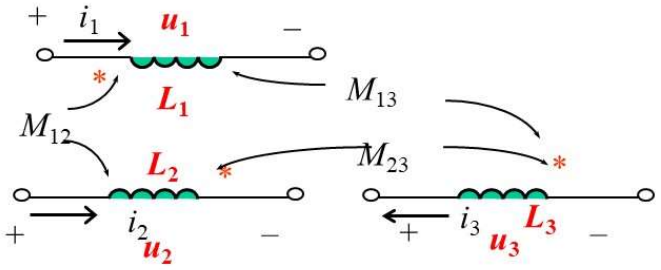
如图标注的同名端是



A 正确的

B 错误的

Ex：标出三个有互感耦合线圈的自感电压及互感电压，并列出其KVL方程。

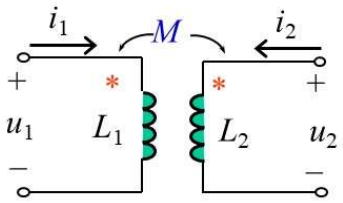


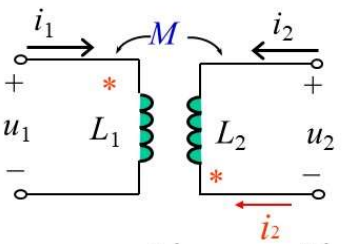
$$u_1 = L_1\frac{di_1}{dt} - M_{12}\frac{di_2}{dt} + M_{13}\frac{di_3}{dt}$$

$$u_2 = L_2\frac{di_2}{dt} - M_{12}\frac{di_1}{dt} - M_{23}\frac{di_3}{dt}$$

$$u_3 = -L_3\frac{di_3}{dt} - M_{13}\frac{di_1}{dt} + M_{23}\frac{di_2}{dt}$$

例3





时域形式

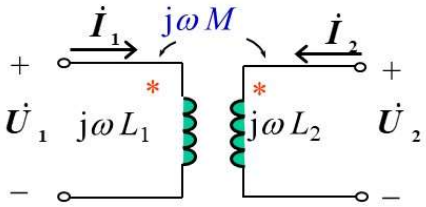
$$u_1 = L_1\frac{\mathrm{d}i_1}{\mathrm{d}t} + M\frac{\mathrm{d}i_2}{\mathrm{d}t}$$

$$u_2 = M\frac{\mathrm{d}i_1}{\mathrm{d}t} + L_2\frac{\mathrm{d}i_2}{\mathrm{d}t}$$

$$u_1 = L_1\frac{\mathrm{d}i_1}{\mathrm{d}t} + M\frac{\mathrm{d}i_2}{\mathrm{d}t}$$

$$u_2 = -M\frac{\mathrm{d}i_1}{\mathrm{d}t} - L_2\frac{\mathrm{d}i_2}{\mathrm{d}t}$$

在正弦稳态分析中，其相量形式的方程为

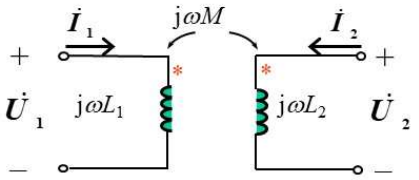


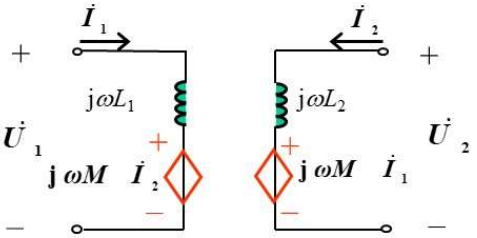
$$\dot{U}_1 = j\omega L_1\dot{I}_1 + j\omega M\dot{I}_2$$

$$\dot{U}_2 = j\omega M\dot{I}_1 + j\omega L_2\dot{I}_2$$

雨课堂 Rain Classroom

互感线圈的等效电路

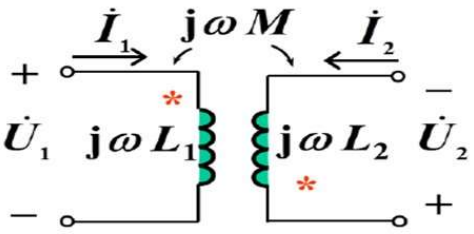




$$\begin{cases} \dot{U}_1 = \mathrm{j}\omega L_1 \dot{I}_1 + \mathrm{j}\omega M \dot{I}_2 \\ \dot{U}_2 = \mathrm{j}\omega L_2 \dot{I}_2 + \mathrm{j}\omega M \dot{I}_1 \end{cases}$$

单选题 1分

下列公式正确的是
“ ？ ”



A
$$\dot{U}_1 = \mathrm{j}\omega L_1 \dot{I}_1 + \mathrm{j}\omega M \dot{I}_2$$
$$\dot{U}_2 = \mathrm{j}\omega M \dot{I}_1 + \mathrm{j}\omega L_2 \dot{I}_2$$

B
$$\dot{U}_1 = \mathrm{j}\omega L_1 \dot{I}_1 - \mathrm{j}\omega M \dot{I}_2$$
$$\dot{U}_2 = \mathrm{j}\omega M \dot{I}_1 + \mathrm{j}\omega L_2 \dot{I}_2$$

C
$$\dot{U}_1 = \mathrm{j}\omega L_1 \dot{I}_1 - \mathrm{j}\omega M \dot{I}_2$$
$$\dot{U}_2 = \mathrm{j}\omega M \dot{I}_1 - \mathrm{j}\omega L_2 \dot{I}_2$$

D
$$\dot{U}_1 = -\mathrm{j}\omega L_1 \dot{I}_1 - \mathrm{j}\omega M \dot{I}_2$$
$$\dot{U}_2 = -\mathrm{j}\omega M \dot{I}_1 + \mathrm{j}\omega L_2 \dot{I}_2$$

四 耦合系数k (coupling coefficient)

$$k \overset{\text{def}}{=} \frac{M}{\sqrt{L_1 L_2}} \qquad M^2 \le L_1 L_2 \Longrightarrow k \le 1$$

互感不大于两个自感的几何平均值。

全耦合: $k = 1 \Longrightarrow \Phi_{S1} = \Phi_{S2} = 0$

互感现象
- 利用——变压器，信号和功率的传递
- 避免——干扰

克服：合理布置线圈相互位置减少互感作用



$k<1$　　$k=1$　　$k=0$

单选题 1分

对于两个耦合的电感线圈，假定其电感分别为**2mH**和**8mH**，两者间可能的最大互感为

A　2mH

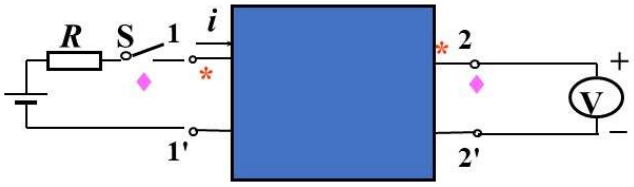
B　4mH

C　6mH

D　8mH

五、同名端的实验测定

当随时间增大的电流从一线圈的一端流入时，将会引起另一线圈相应同名端的电位升高。



方法一：

如图电路，当闭合开关S时，i增加，

$\frac{\mathrm{d}i}{\mathrm{d}t} > 0$，　若电压表正偏，则1、2是同名端。

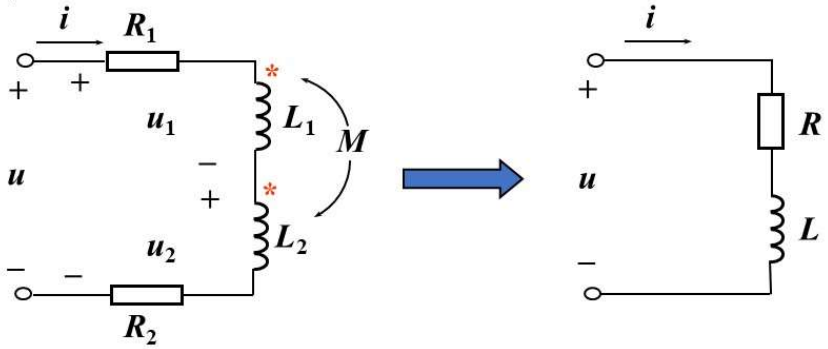
方法二：

输入端接正弦波，用示波器观察输入输出波形的相位关系。

8-2 含耦合电感电路的计算

1. 互感线圈的串联

(1) 顺串

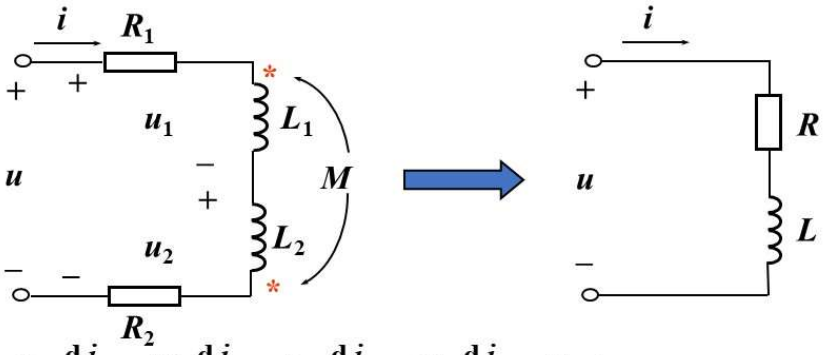


$$u = R_1 i + L_1 \frac{\mathrm{d}i}{\mathrm{d}t} + M \frac{\mathrm{d}i}{\mathrm{d}t} + L_2 \frac{\mathrm{d}i}{\mathrm{d}t} + M \frac{\mathrm{d}i}{\mathrm{d}t} + R_2 i$$

$$= (R_1 + R_2)i + (L_1 + L_2 + 2M)\frac{\mathrm{d}i}{\mathrm{d}t} = Ri + L\frac{\mathrm{d}i}{\mathrm{d}t}$$

$$\therefore \quad R = R_1 + R_2 \qquad L = L_1 + L_2 + 2M$$

(2) 反串



$$u = R_1 i + L_1 \frac{\mathrm{d}i}{\mathrm{d}t} - M \frac{\mathrm{d}i}{\mathrm{d}t} + L_2 \frac{\mathrm{d}i}{\mathrm{d}t} - M \frac{\mathrm{d}i}{\mathrm{d}t} + R_2 i$$

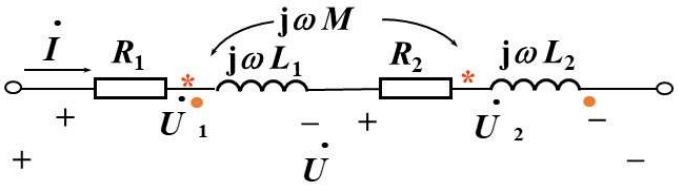
$$= (R_1 + R_2) i + (L_1 + L_2 - 2M) \frac{\mathrm{d}i}{\mathrm{d}t} = Ri + L \frac{\mathrm{d}i}{\mathrm{d}t}$$

$$\therefore \quad R = R_1 + R_2 \qquad L = L_1 + L_2 - 2M$$

$$L = L_1 + L_2 - 2M \geq 0 \qquad \therefore \ M \leq \frac{1}{2}(L_1 + L_2)$$

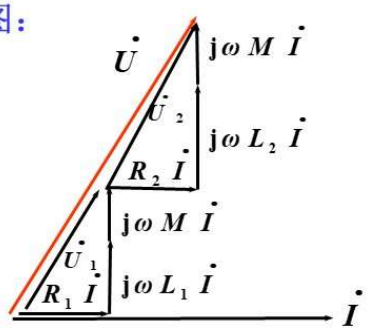
互感不大于两个自感的算术平均值。

在正弦激励下：

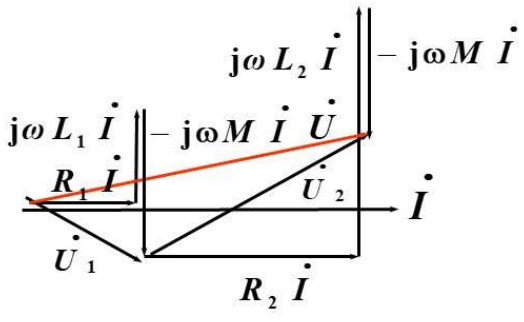


$$\dot{U} = (R_1 + R_2)\dot{I} + \mathrm{j}\omega(L_1 + L_2 \pm 2M)\dot{I}$$

相量图：



(a) 正串



(b) 反串

问题：如何测量互感值？

$$L_{顺}=L_1+L_2+2M \qquad L_{反}=L_1+L_2-2M$$

* 顺接一次，反接一次，就可以测出互感：

$$M=\frac{L_{顺}-L_{反}}{4}$$

* 全耦合 $M=\sqrt{L_1L_2}$

当 $L_1=L_2=L$ 时，$M=L$

$$L_{eq}=\begin{cases}4M & \text{顺串}\\ 0 & \text{反串}\end{cases}$$

2023年5月6日



单选题 1分

两电感线圈同名端顺串连接时电感值为10mH，同名端反串连接时电感值为2mH，则其互感为

A 8 mH

B 2 mH

C 4 mH

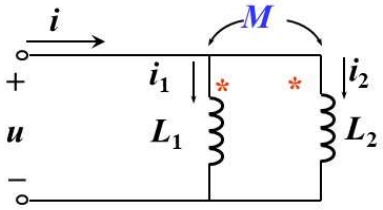
D 5 mH



雨课堂 Rain Classroom

2. 互感线圈的并联

(1) 同名端在同侧



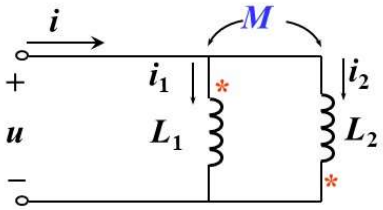
$$\begin{cases} u = L_1 \dfrac{\mathrm{d}i_1}{\mathrm{d}t} + M \dfrac{\mathrm{d}i_2}{\mathrm{d}t} \\ u = L_2 \dfrac{\mathrm{d}i_2}{\mathrm{d}t} + M \dfrac{\mathrm{d}i_1}{\mathrm{d}t} \\ i = i_1 + i_2 \end{cases}$$

解得u, i的关系：

$$u = \frac{(L_1 L_2 - M^2)}{L_1 + L_2 - 2M} \frac{\mathrm{d}i}{\mathrm{d}t}$$

$$L_{\mathrm{eq}} = \frac{(L_1 L_2 - M^2)}{L_1 + L_2 - 2M} \geq 0$$

(2) 同名端在异侧



$$\begin{cases} u = L_1 \dfrac{\mathrm{d}i_1}{\mathrm{d}t} - M \dfrac{\mathrm{d}i_2}{\mathrm{d}t} \\ u = L_2 \dfrac{\mathrm{d}i_2}{\mathrm{d}t} - M \dfrac{\mathrm{d}i_1}{\mathrm{d}t} \\ i = i_1 + i_2 \end{cases}$$

解得u, i的关系：

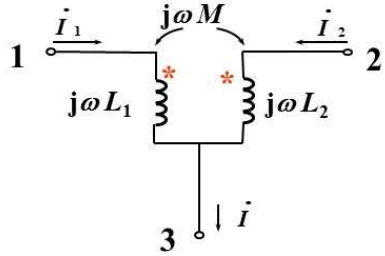
$$u = \frac{(L_1 L_2 - M^2)}{L_1 + L_2 + 2M} \frac{\mathrm{d}i}{\mathrm{d}t}$$

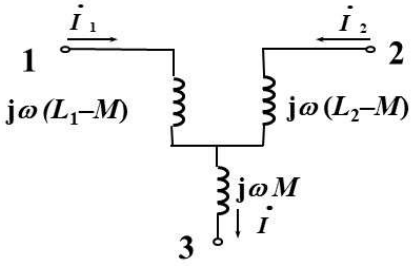
$$L_{\mathrm{eq}} = \frac{(L_1 L_2 - M^2)}{L_1 + L_2 + 2M} \geq 0$$

雨课堂 Rain Classroom

3. 有一个公共节点互感线圈的去耦等效电路

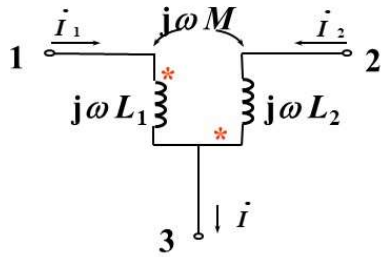
(1) 同名端同侧联接

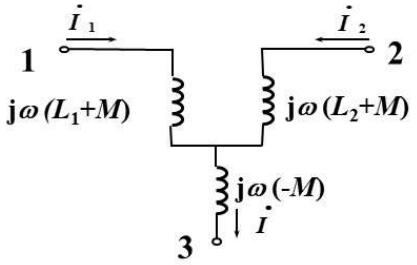




$$\begin{cases} \dot{U}_{13} = \mathrm{j}\omega L_1 \dot{I}_1 + \mathrm{j}\omega M \dot{I}_2 \\ \dot{U}_{23} = \mathrm{j}\omega L_2 \dot{I}_2 + \mathrm{j}\omega M \dot{I}_1 \\ \dot{I} = \dot{I}_1 + \dot{I}_2 \end{cases} \quad \text{整理得} \quad \begin{cases} \dot{U}_{13} = \mathrm{j}\omega(L_1 - M)\dot{I}_1 + \mathrm{j}\omega M \dot{I} \\ \dot{U}_{23} = \mathrm{j}\omega(L_2 - M)\dot{I}_2 + \mathrm{j}\omega M \dot{I} \\ \dot{I} = \dot{I}_1 + \dot{I}_2 \end{cases}$$

(2) 同名端异侧联接



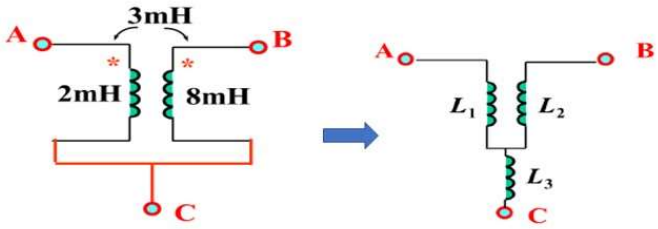


$$\begin{cases} \dot{U}_{13} = \mathrm{j}\omega L_1 \dot{I}_1 - \mathrm{j}\omega M \dot{I}_2 \\ \dot{U}_{23} = \mathrm{j}\omega L_2 \dot{I}_2 - \mathrm{j}\omega M \dot{I}_1 \\ \dot{I} = \dot{I}_1 + \dot{I}_2 \end{cases} \quad \text{整理得} \quad \begin{cases} \dot{U}_{13} = \mathrm{j}\omega(L_1 + M)\dot{I}_1 - \mathrm{j}\omega M \dot{I} \\ \dot{U}_{23} = \mathrm{j}\omega(L_2 + M)\dot{I}_2 - \mathrm{j}\omega M \dot{I} \\ \dot{I} = \dot{I}_1 + \dot{I}_2 \end{cases}$$

雨课堂 Rain Classroom

单选题 1分

如图所示，去耦等效电路中，L_1的电感值为



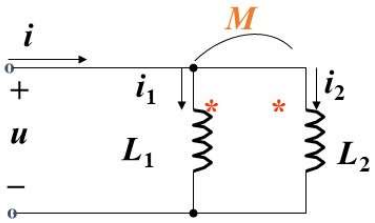
A 1mH

B −1mH

C 5mH

D 3mH

并联电路的去耦等效分析



$$u = L_1 \frac{\mathrm{d}i_1}{\mathrm{d}t} + M \frac{\mathrm{d}i_2}{\mathrm{d}t}$$

$$= (L_1 - M)\frac{\mathrm{d}i_1}{\mathrm{d}t} + M \frac{\mathrm{d}i}{\mathrm{d}t}$$

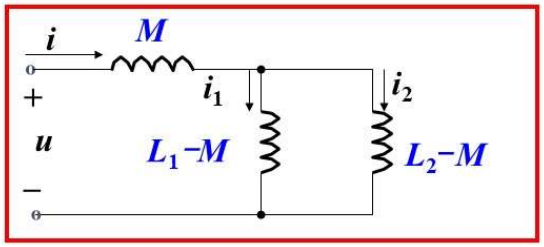
$i_2 = i - i_1$

$$u = L_2 \frac{\mathrm{d}i_2}{\mathrm{d}t} + M \frac{\mathrm{d}i_1}{\mathrm{d}t}$$

$$= (L_2 - M)\frac{\mathrm{d}i_2}{\mathrm{d}t} + M \frac{\mathrm{d}i}{\mathrm{d}t}$$

$i_1 = i - i_2$

$$L_{eq} = \frac{(L_1 L_2 - M^2)}{L_1 + L_2 - 2M}$$ 等效电路



$(L_1-M)//(L_2-M)+M$